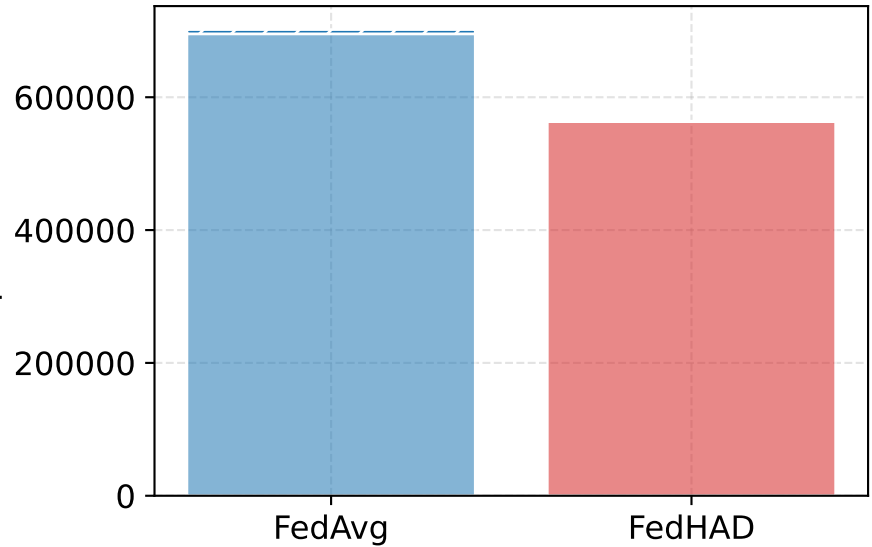


Productive ($\cos_sim \geq 0.2$) Ineffective ($\cos_sim < 0.2$)

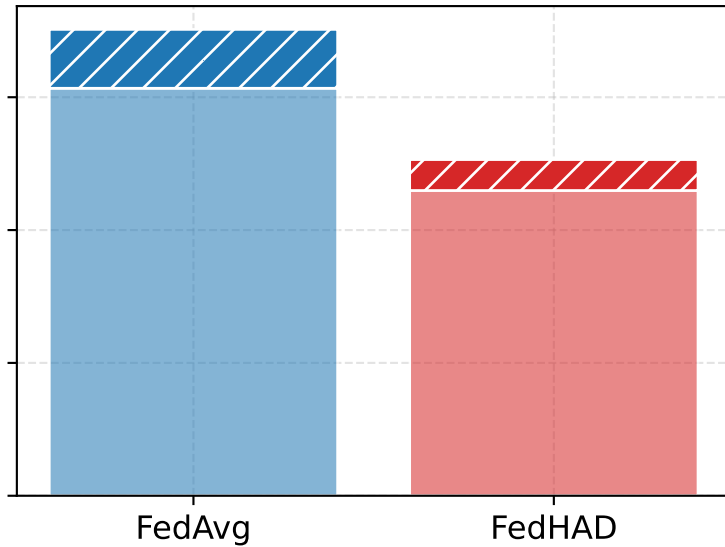
Minibatch steps (total, 10 seeds)

$\alpha=1.0$



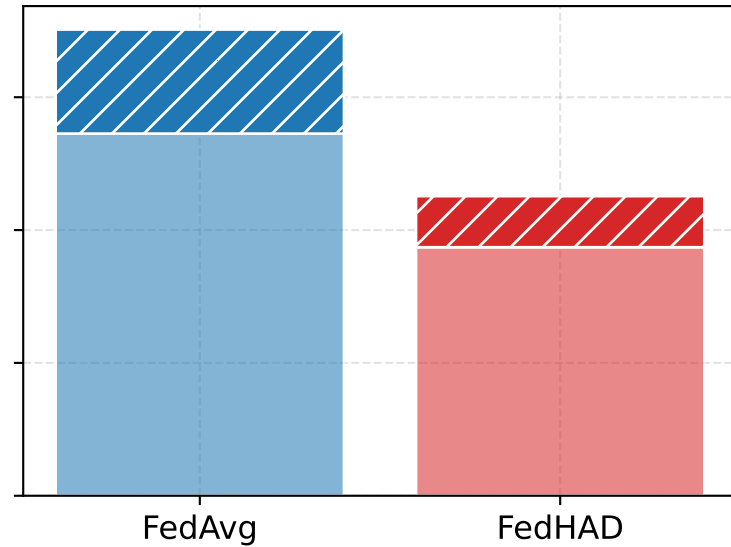
n ineffective:
FedAvg=6,390
FedHAD=552
(small counts —
unstable %)

$\alpha=0.1$



n ineffective:
FedAvg=88,585
FedHAD=46,528

$\alpha=0.01$



n ineffective:
FedAvg=156,670
FedHAD=76,925